



Beyond Binary Metrics: Unveiling the Safety Illusion in Autonomous Driving Simulation

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Binary Metrics Count Collisions. *CCM Measures Consequences.*

1. MOTIVATION / PROBLEM

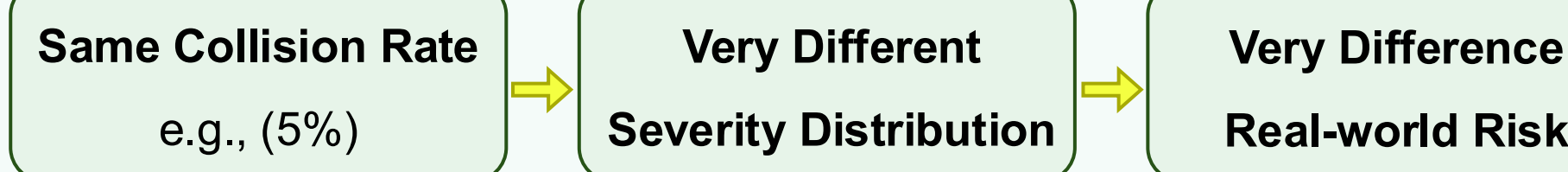
- Binary collision labels treat all crashes equally.
- They ignore crash severity and hidden long-tail risk.
- Minor grazes $\neq$ fatal crashes.
- As a result, models can game the metric by reducing minor contacts while missing rare, **high-severity events**.



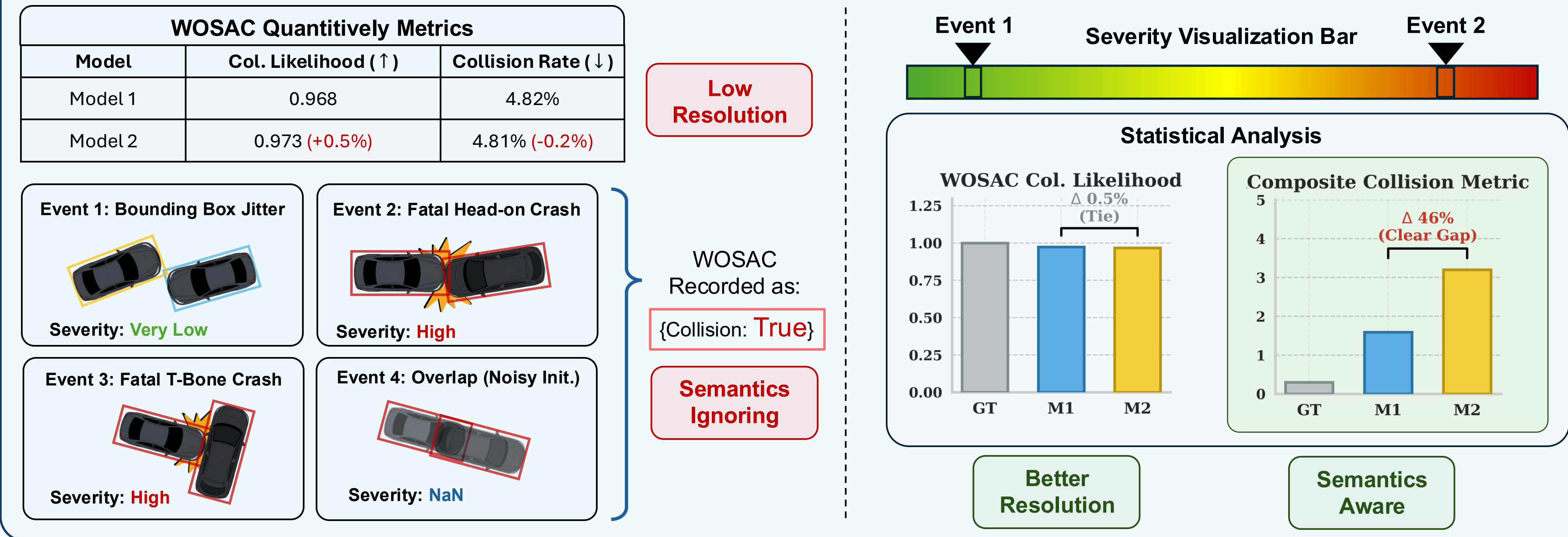
We need a metric that reflects **CONSEQUENCES**, not just **OCCURRENCES**.

2. WHY COLLISION RATE IS NOT ENOUGH

- Binary metrics ignore severity differences.
- Many small contacts can inflate collision rate.
- Rare catastrophic events are undercounted.
- Poor correlation with real-world risk.



3. WOSAC VS. COMPOSITE COLLISION METRIC (CCM)



4. METHOD OVERVIEW

Coposite Severity of an Event

$$S(\text{Event}) = m(v_{rel}) * \delta(d) * g(\Delta_t)$$

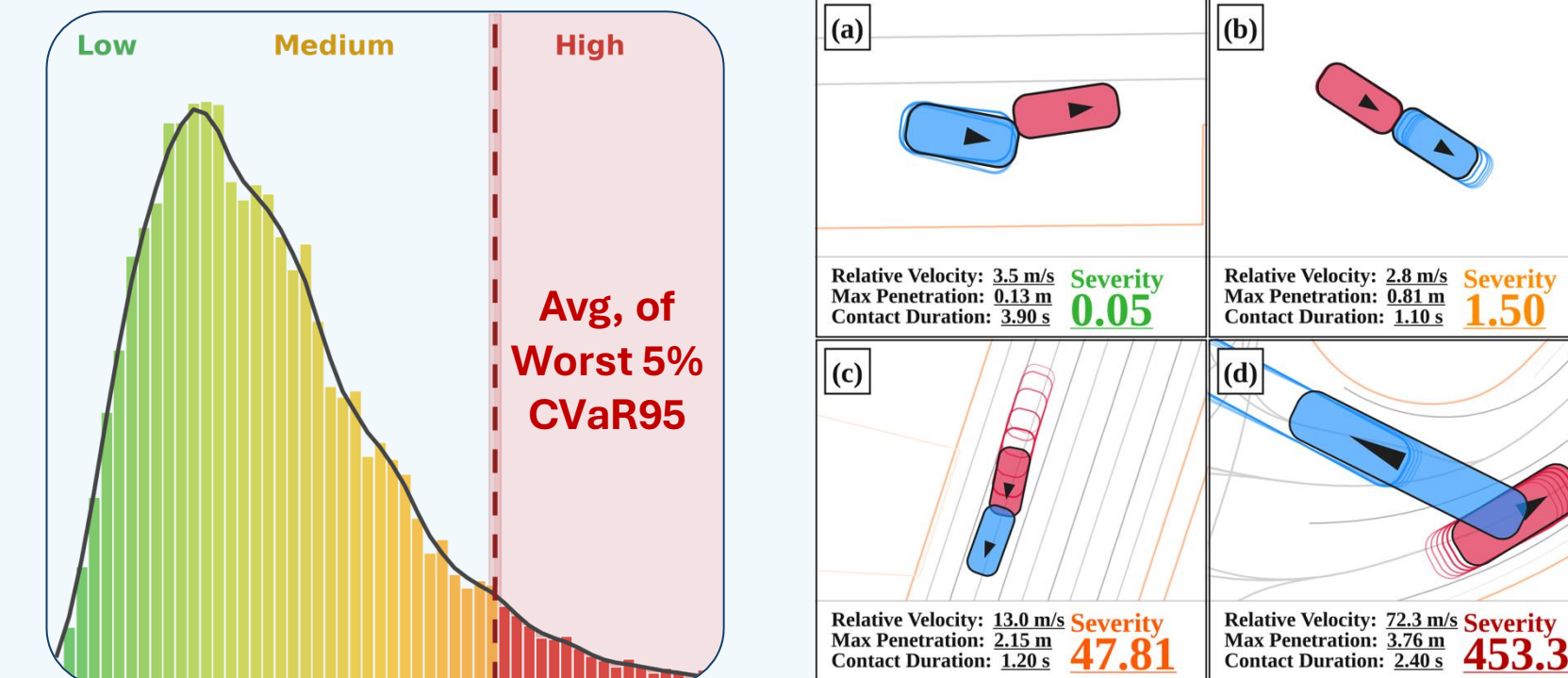
Velocity $m(v_{rel})$	Higher Relative Speed $\rightarrow$ Higher Severity
Depth $\delta(d)$	More Overlap $\rightarrow$ Higher Severity
Duration $g(\Delta_t)$	Longer Interaction $\rightarrow$ Higher Severity

We aggregate event severity using tail risk **CVaR95** to capture rare but critical failures.

5. COMPOSITE COLLISION METRIC (CCM)

$$CCM = CVaR_{95}(S(event))$$

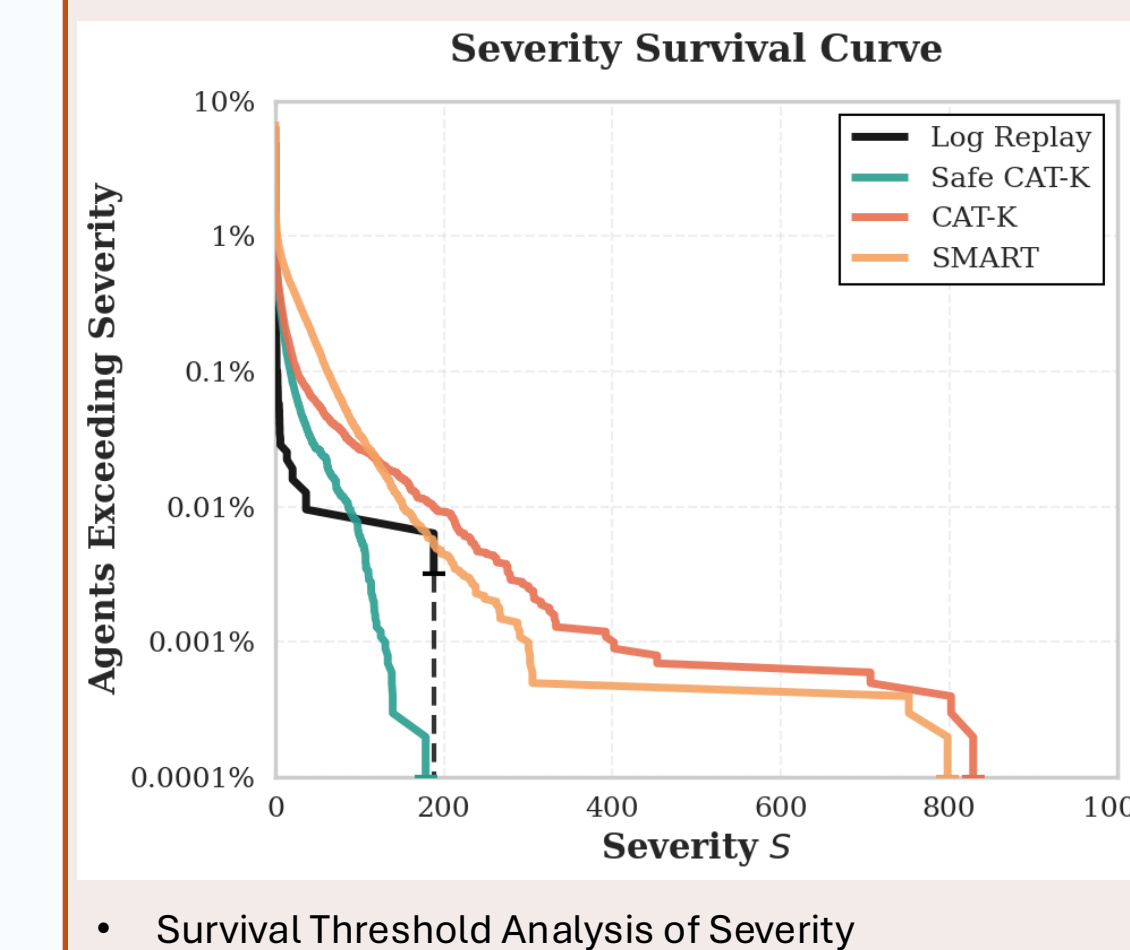
- Uses Conditional Value-at-Risk at 95%.
- Focuses on the worst 5% tail of events.



Interpretation:

Lower CCM $\rightarrow$ safer behavior with fewer severe outcomes.

6. EXPERIMENT & KEY RESULTS



Model	WOSAC Col. Likelihood	Origin Col. Rate (%)	Inter-Vehicle Col. Rate (%)	CCM
Log Replay	—	4.69	0.72	0.028
SMART	0.9643	6.75	2.36	5.557
CAT-K	0.9672	6.16	1.62	2.792
Safe CAT-K	0.9637	6.23	1.51	1.611

- Safety related Rejection-Sampling is applied to **Safe CAT-K**

WOSAC (Likelihood):

- Similar scores for different failure modes.

CCM (CVaR95):

- Clear separation of physical risk.

CCM reflects **CONSEQUENCES**, not just **OCCURRENCES**.



TAKEAWAY / CONCLUSION



High benchmark realism does not guarantee physical safety.



Current sim-agents still produce severe long-tail collisions.



Severity-aware evaluation is essential for useful autonomous driving simulation.